Alzheimer Features

National Central University

Statistical Practice

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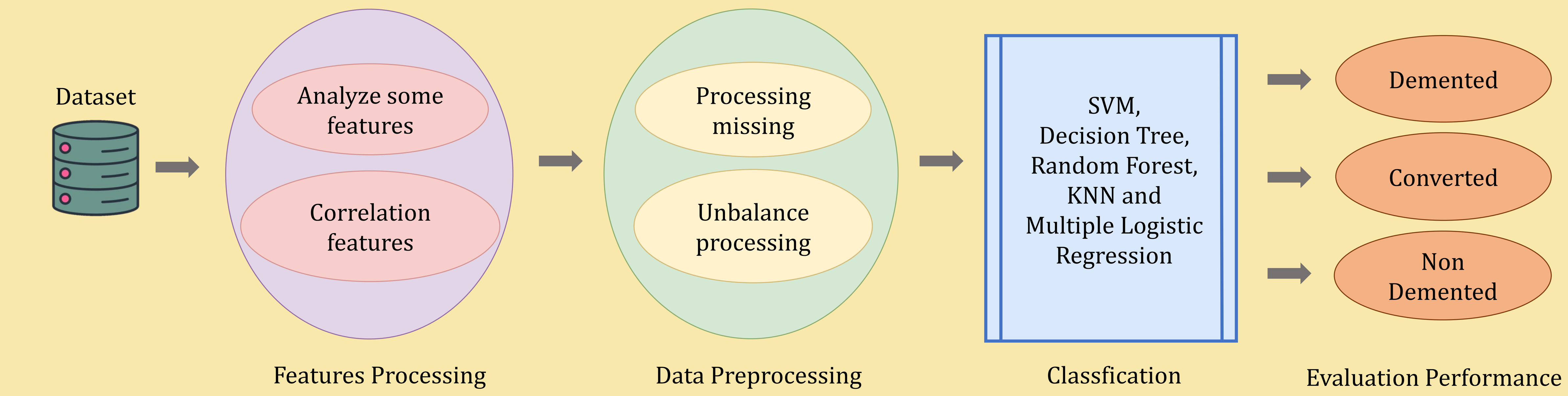
Abstract

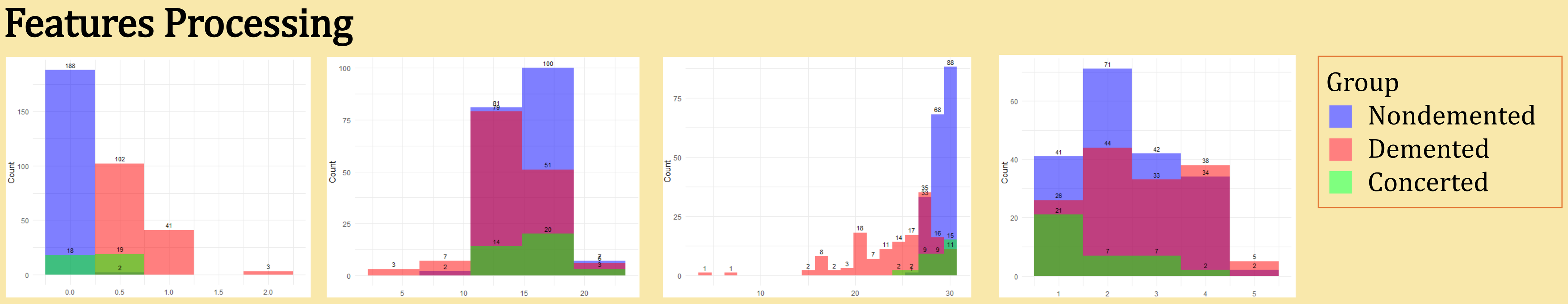
Dementia and Alzheimer's disease are caused by neurodegeneration and poor communication between neurons in the brain. With no effective treatments currently available, early diagnosis is critical to prevent disease progression. This study explores the use of machine learning models such as SVM, Decision Tree, Random Forest, KNN, and Logistic Regression to classify individuals into three categories: Demented, Converted, and Non-Demented. Feature analysis and data preprocessing techniques, including handling missing data and addressing class imbalance, were applied to enhance model performance. Results demonstrate the effectiveness of preprocessing and the potential of SVM in achieving accurate predictions.

Motivation

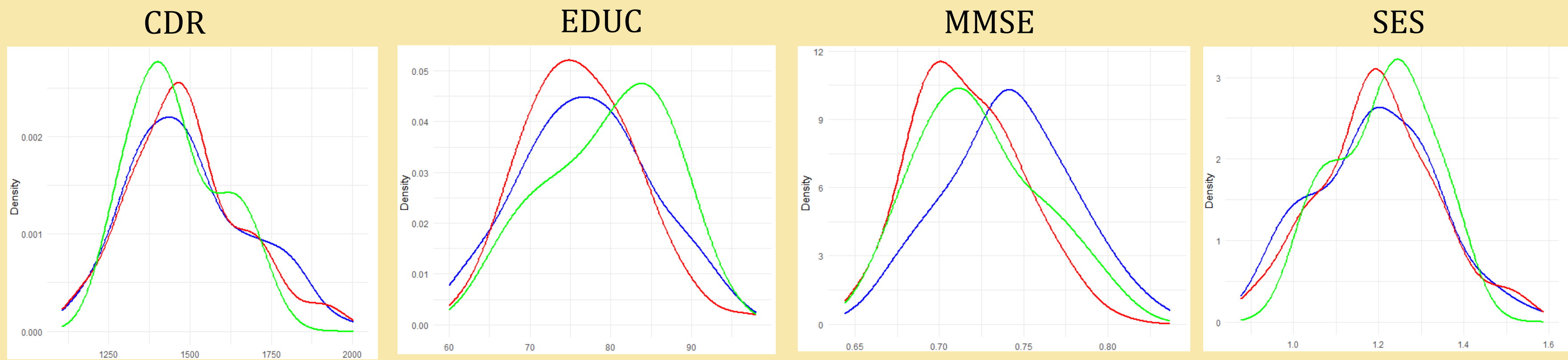
The elderly population in Taiwan is steadily increasing, and with population aging, the number of dementia cases is likely to rise. Therefore, we believe that understanding the features of Alzheimer's disease is crucial. This knowledge not only facilitates early diagnosis and intervention but also advances prevention and treatment, ultimately reducing social and economic burdens.



Features Processing



CDR EDUC MMSE SES



eTIV Age nWBV ASF

CDR → 臨床痴呆評分

MMSE → 簡易精神狀態檢查

SES → 社會經濟地

eTIV → 估計總顱內體積

nWBV → 標準化全腦體積

ASF → 大腦影像比例因子

Data Preprocessing

Processing missing

Our data has missing values in SES and MMSE, we are using the median to fill those missing values.

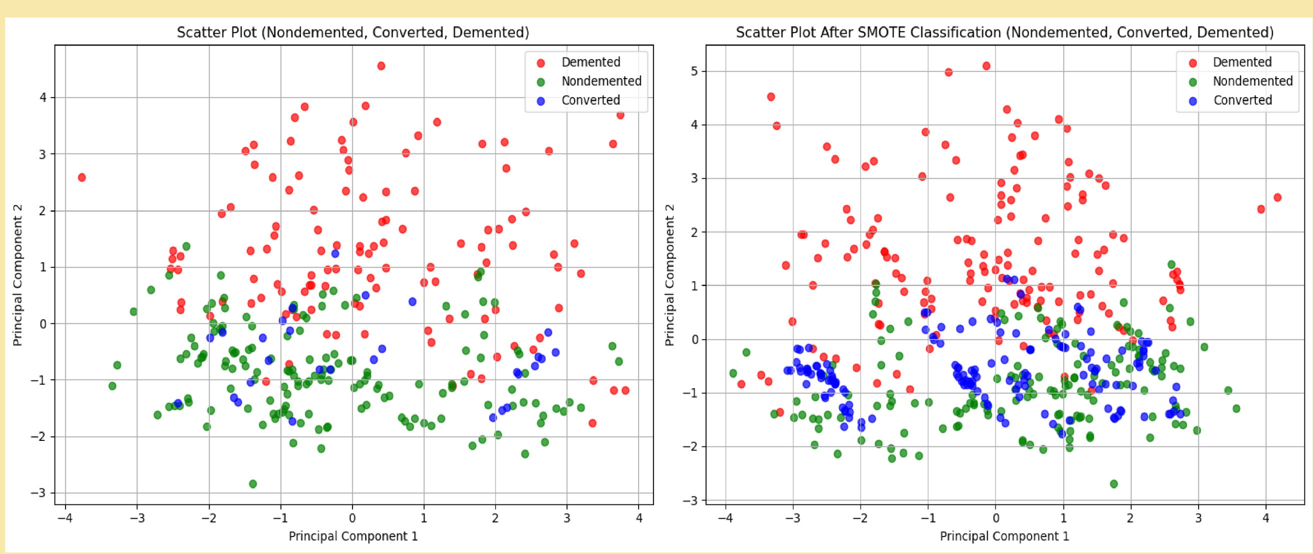
Unbalance processing

We can observe that the number of samples in the Converted class is significantly lower, indicating that the dataset has a class imbalance problem. This may affect the performance of the model.

OASIS Dataset					
Training 80%			Testing 20%		
Non-Demented	Demented	Converted	Non-Demented	Demented	Converted
152	117	30	38	29	7

Three-Class

Due to the issue of data imbalance, we used the SMOTE method to increase the sample size, bringing the number of samples in each of the three groups to 152. We also applied 10-fold cross-validation to fine-tune the model parameters and check for potential overfitting. It was observed that, except for the KNN method, the other methods performed quite well, with SVM achieving the highest prediction accuracy. However, it is worth noting that the precision for the Converted group was particularly low across all methods. Moving forward, we plan to explore other approaches to address the data imbalance issue.



before after

Data prepocessing (SMOTE)

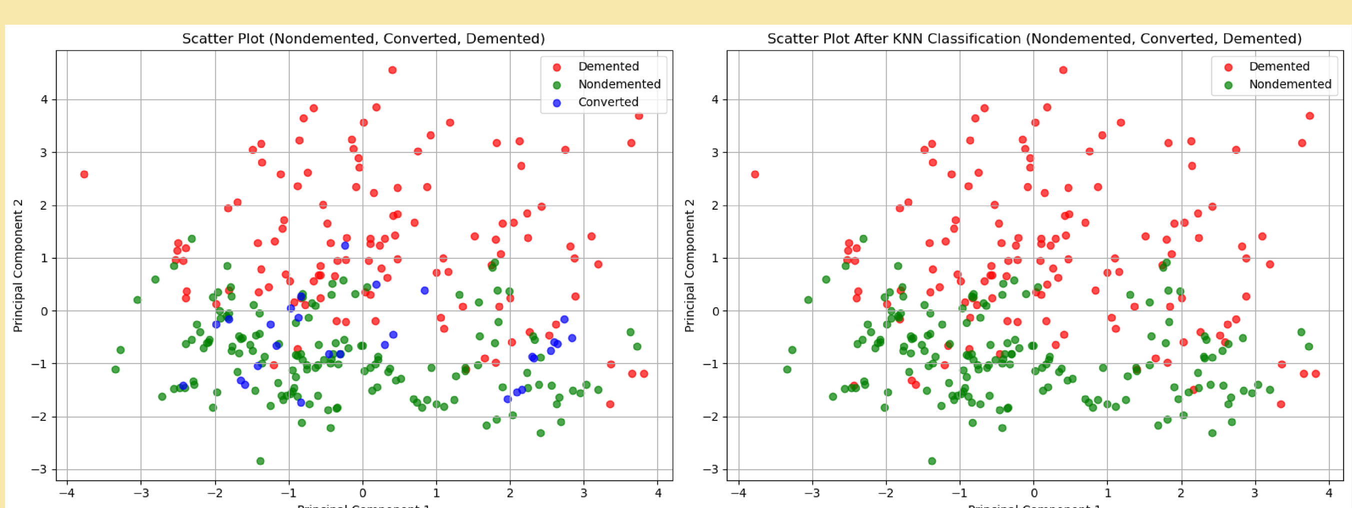
	SVM			Decision Tree			Random Forest			KNN			Multiple Logistic Regression		
class	N	C	D	N	C	D	N	C	D	N	C	D	N	C	D
10-fold CV %		96.05			90.16			91.68			75.47			89.01	
Accuracy %		90.67			88			80			57.33			89.33	
Precision %	94.74	37.5	100	97.37	12.5	96.55	92.11	37.5	75.86	57.89	50	58.62	97.37	25	96.55
Recall %	97.3	75	85.29	92.5	50	84.85	79.55	60	84.62	73.33	21.05	65.38	76.92	66.67	84.85
F1 score %	96	50	92.06	94.87	20	90.32	85.37	46.15	80	64.71	29.63	61.82	77.92	36.36	90.32

N：Non-Dementia, C：Converted, D：Dementia

Classfications

Two-Class

We used the KNN method to redistribute the 30 Converted samples into the other two groups, increasing Non-Dementia to 167 samples and Dementia to 132 samples. We applied the following four models for prediction. The models with the highest accuracy were SVM and Decision Tree, both achieving an accuracy of 96%. The lowest accuracy was still observed with the KNN model. However, the use of KNN for reclassification improved the overall prediction accuracy without causing issues with low precision. Therefore, we believe this method is more suitable.



before after

Data prepocessing (KNN)

	SVM		Decision Tree		Random Forest		KNN		Logistic Regression	
class	N	D	N	D	N	D	N	D	N	D
10-fold CV %	94.32		93.98		87.65		76.21		92.98	
Accuracy %	96		96		84		66.67		94.67	
Precision %	100	91.67	97.44	94.44	92.31	75	66.67	66.67	100	88.89
Recall %	92.86	100	95	97.14	80	90	68.42	64.86	90.7	100
F1 score %	96.30	95.65	96.20	95.77	85.71	81.82	67.53	65.75	95.12	94.12

N：Non-Dementia, D：Dementia

Classfications

Conclusion

For three-class classification, Support Vector Machine (SVM) demonstrated the best overall performance when data was preprocessed with SMOTE. It is well-suited for predicting the categories of Non-Dementia, Converted, and Dementia. However, there is an issue with relatively low precision.

For two-class classification, SVM and Decision Tree performed exceptionally well after KNN preprocessing, particularly for distinguishing between Non-Dementia and Dementia.

References

[1]Khan, A., Zubair, S., & Khan, S. (2022). A systematic analysis of assorted machine learning classifiers to assess their potential in accurate prediction of dementia. *Arab Gulf Journal of Scientific Research, 40*(1)

[2]Mohammed, B. A., Senan, E. M., Rassem, T. H., Makbol, N. M., Alanazi, A. A., Al-Mekhlafi, Z. G., Almurayziq, T. S., & Ghaleb, F. A. (2022). Multi-method analysis of medical records and MRI images for early diagnosis of dementia and Alzheimer's disease based on deep learning and hybrid methods..